



EN – 019

V Semester B.Tech. (CSE and AIML) Degree
Examination, July/August 2024
(NEP Scheme)

21CSPC501 : PROBABILITY AND STOCHASTIC PROCESSES

Time : 3 Hours

Max. Marks : 100

Instruction : Answer **any five** full questions choosing atleast **one** full question from **each Unit (1 or 2, 3 or 4, 5 or 6, 7 or 8, 9 or 10)**

1. a) A box contains 5 green, 8 blue and 6 red balls. A ball is chosen at random from the box. A second ball is chosen after replacing the first ball. What is the probability of choosing a green and then a red ball ? **10**
- b) The probability of winning a cricket tournament by India against Pakistan is 0.4, 0.6 against Srilanka and 0.8 against Bangladesh. India plays a game against a randomly chosen opponent. What is the probability of winning ? **10**

OR

2. a) The exam scores of 7 students are : 65, 70, 75, 80, 85, 90, 95. Calculate :
i) The mean of the scores.
ii) The median of the scores.
iii) If an additional student scored 100, how would the mean and median change ? **10**
- b) In a class of 60 students, 10% of the male students have scored a CGPA of over 9.0 and 5% of the female students have scored a CGPA 9.0. The total student population is divided in the male : female ratio of 3 : 2. If a student among all those who have scored CGPA of over 9.0 is chosen at random, what is the probability that the chosen student is a female ? **10**
3. a) Consider tossing a coin three times. Let the random variable X denote the number of heads that come up. Determine the cumulative distribution function $F_X(x)$. **10**

- b) A random variable has the following probability distribution function :

x	0	1	2	3	4	5	6	7
P(X = x)	0	k	2k	2k	3k	k ²	2k ²	7k ² +k

Find :

- i) The value of k
ii) Evaluate $P(X < 6)$
iii) Evaluate $P(X \geq 6)$
iv) If , $P(X \leq c) > \frac{1}{2}$ then find the minimum value of c. **10**

OR

P.T.O.



4. a) The amount of time in hours that an electric bulb functions before breaking down is a continuous random variable with pdf given by :

$$f(x) = \begin{cases} \lambda e^{\frac{-x}{100}} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

What is the probability that

i) The bulb will function between 200 to 300 hours before breaking down and

ii) It will function for less than 250 hours ? 10

- b) Suppose that X is a continuous random variable with pdf given by :

$$f_X(x) = \begin{cases} 2x & x \in (0, 1) \\ 0 & \text{Otherwise} \end{cases}$$

Find the expectation and variance of X. 10

5. a) A bulb manufacturing company is known to produce 5% defective. In a random sample of 15 bulbs, what is the probability that there are :

i) Exactly 3 defectives and

ii) Not more than 3 defectives. 10

- b) An urn contains N white and M black balls. Balls are randomly selected, one at a time, until a black one is obtained. If we assume that each selected ball is replaced before the next one is drawn, what is the probability that :

i) Exactly n draws are needed and

ii) At least k draws are needed ? 10

OR

6. a) Suppose we're interested in modeling the time until the next customer arrives at a store, and we know that customers arrive at an average rate of 4 customers per hour. We can use the exponential distribution to model the time between successive customer arrivals. 10

- b) Arrivals of customers at a certain checkout counter follow a Poisson distribution. It is known that, during a given 30-minute period, one customer arrived at the counter. Find the probability that the customer arrived during the last 5 minutes of the 30-minute period. 10



7. a) Suppose that the probability of a dry day following a rainy day is $1/3$ and the probability of a rainy day following a dry day is $1/2$. Given that May 1 is a dry day. Find the probability that May 3 is a dry day and also May 5 is a dry day. **10**

- b) Consider a simple Markov chain with three states and the following transition probability matrix P :

$$P = \begin{bmatrix} 0.2 & 0.5 & 0.3 \\ 0.4 & 0.4 & 0.2 \\ 0.3 & 0.3 & 0.4 \end{bmatrix}$$

Find the steady state probability vector $\pi = [\pi_1, \pi_2, \pi_3]$.

10

OR

8. a) Consider a CTMC with three states 0,1,2 and the following generator matrix Q :

$$Q = \begin{bmatrix} -0.5 & 0.3 & 0.2 \\ 0.5 & -0.8 & 0.3 \\ 0.4 & 0.4 & -0.8 \end{bmatrix}$$

Find the transition probability matrix $P(t)$ at time t , solve either the Kol-Mogorov forward or backward equations using the Chapman-Kol-Mogorov relation.

10

- b) Generate a generator matrix (Q) for a Birth : Death process. Derive the balance equations for $n = 0$ and $n \geq 1$, where n is the number of state. **10**

9. a) Consider a call center with 3 agents. Customers arrive at the call center according to a Poisson process with a rate of 30 customers per hour. Each agent can handle calls at an average rate of 15 customers per hour.

- i) Calculate the traffic intensity (ρ) of the system.
- ii) Determine the probability π_0 that there are no customers in the system (i.e., all agents are idle).
- iii) Determine the steady-state probabilities that there are exactly 0,1,2,3,4, and 5 customers in the system.
- iv) Compute the average number of customers in the system (L).
- v) Compute the average number of customers waiting in the queue L_q .
- vi) Calculate the average time a customer spends in the system (W).
- vii) Calculate the average time a customer spends waiting in the queue W_q . **10**



- b) Consider a network of queues representing a customer service system with three service nodes : A, B, and C. Customers arrive at node A, can move to node B for additional assistance, and then proceed to node C for final resolution. The average arrival rate of customers at node A is 10 customers per hour. On average, 80% of customers at node A move to node B, while the remaining customers leave the system after node A. node B has a service rate of 15 customers per hour, and 30% of customers at node B proceed to node C. At node C, the service rate is 20 customers per hour. Calculate the following metrics :

- i) Arrival rates at node A, B and C.
- ii) Service rates at node B and C.
- iii) Utilization and
- iv) System throughput.

10

OR

10. a) Consider a closed queuing network with three service points : A, B and C. The average arrival rate of customers entering the system is 20 customers per hour. At service point A, the service rate is 15 customers per hour, and 80% of customers move to service point B while the remaining 20% exit the system. At point B, the service rate is 12 customers per hour, and 70% of customers move to point C. At point C, the service rate is 18 customers per hour, and all customers complete service and return to point A. Calculate :

- i) The average number of customers at each service point.
- ii) The average time a customer spends in the system.
- iii) The system throughput.

10

- b) Consider a customer service centre where customers arrive to request assistance with varying service times that do not follow an exponential distribution. Discuss the impact on the following performance metrics.

- i) Variability
- ii) Queue dynamics
- iii) Average waiting time
- iv) Resource utilization

10



EN – 020

**V Semester B.Tech. (CSE and AIML) Degree Examination, July/August 2024
(NEP Scheme)
21AIPC501 : GAME THEORY**

Time : 3 Hours

Max. Marks : 100

Instruction : Answer **any five full** questions by selecting one from **each** Unit.

UNIT – I

1. a) Define Strategic Game. How can a two-player strategic game, where each player has two actions, be conveniently represented ? **8**
- b) What are the key elements that constitute a model of rational choice when there is no uncertainty ? **4**
- c) Find all Nash equilibria of the following games and the dominant strategy, if available. **8**

i)

	L	R
U	2, 4	1, 3
D	3, 1	5, 3

ii)

	S	M	L
S	6, 4	4, 6	1, 0
M	2, 3	5, 3	3, 6
L	12, 9	10, 11	4, 5

OR

2. a) State the lemma and prove that every finite strategic game has a mixed strategy Nash equilibrium. **8**
- b) Define Nash equilibrium of a Bayesian game. **4**

P.T.O.



- c) Consider the following game of incomplete information between Player 1 and Player 2. Player 1's type is known but Player 2 may be either an H type (with probability p) or an L type (with probability $1-p$). The payoffs to this simultaneous game of incomplete information are as follows : Let p be 0.75 or $\frac{3}{4}$.

		Player 2 (Type H)	
		X	Y
Player 1	A	2, 4	2, 3
	B	4, 2	3, 6

		Player 2 (Type L)	
		X	Y
Player 1	A	4, 3	2, 4
	B	3, 2	1, 5

- i) Find all pure strategy Bayes-Nash equilibria under this assumption.
- ii) Find all pure strategy Bayes-Nash equilibria for each value of p (since p is a probability p lies in $[0, 1]$).

8

UNIT – II

3. a) Prove that "Every finite extensive game with perfect information has a subgame perfect equilibrium." 8
- b) Discuss the Chain-Store Game strength and weakness. 8
- c) Define a Machine in strategies as machines. 4

OR

4. a) State and prove the One Deviation Property. 8
- b) Elucidate the Centipede Game with its strengths and weakness. 8
- c) Define the Subgame of the extensive game with perfect information. 4

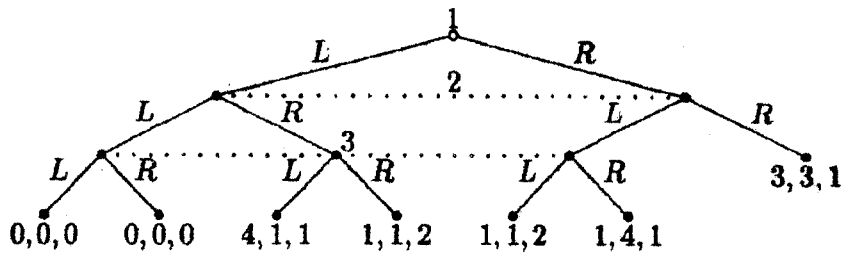
UNIT – III

5. a) Explain Bayesian extensive game and perfect Bayesian Equilibrium. 10
- b) Prove that for any mixed strategy of a player in finite extensive game with perfect recall there is an outcome-equivalent behavioral strategy. 10

OR



6. a) Prove that in the game depicted in Figure below, the unique Nash equilibrium outcome is where players 1 and 2 choose R. Assume player 3's information set is reached with positive probability. Let the strategy profile be β and $\beta_i(I_i)$ (R) = α_i for $i = 1, 2, 3$, where I_i is the single information set of player i . 10



- b) Define Structural Consistency and discuss the components of an Extensive Game with imperfect information. 10

UNIT – IV

7. a) Prove that "Every market with transferable payoff has a non-empty core." And give an example. 10

- b) Demonstrate that the kernel of a coalitional game with transferable payoff is a subset of the bargaining set. 10

OR

8. a) Explain Coalitional game and Exchange Economies. 10

- b) Prove the nucleolus of a coalitional game with transferable payoff is a subset of the kernel. 10

UNIT – V

9. a) Explain the restrictions of bargaining problem. 8

- b) Discuss a symmetric bargaining problem and explain the Symmetry (SYM) axiom in the context of the Nash bargaining solution. Provide an example to illustrate how the SYM axiom ensures fairness between players in a symmetric problem. 10

- c) Define Nash Solution. 2

OR

10. a) Prove that the Nash solution is the only bargaining solution that satisfies PAR, SYM and IIA. 10

- b) Explain why the agreements $x^*(\alpha)$ and $y^*(\alpha)$ proposed in every subgame perfect equilibrium of the bargaining game with a breakdown probability α converge to the Nash solution as $\alpha \rightarrow 0$? What are the key factors and mechanisms driving this convergence and what implications does this have for our understanding of bargaining dynamics under the risk of breakdown? 10



EN – 021

**V Semester B.Tech. (CSE and AIML) Degree Examination, July/August 2024
(NEP Scheme)**

21ISPC501 : DISCRETE MATHEMATICAL STRUCTURES

Time : 3 Hours

Max. Marks : 100

Instruction : Answer **five full** questions.

1. a) Draw a Venn diagram that represents the following relationships for the set A, B, C and D. If $A = \{1, 2, 3, 4, 5\}$, $B = \{1, 2, 3, 4, 5, 6, 8, 9\}$, $C = \{1, 2, 3, 4, 5, 6, 9\}$, $D = \{a, b, c, d, e\}$ then $A \subseteq B$, $A \subseteq C$, $C \not\subseteq D$ and $D \not\subseteq C$. **6**
 - b) What is universal and existential quantification ? Explain with an example. **7**
 - c) Prove that, the following statement is true by using mathematical induction.

$$1^2 + 3^2 + 5^2 + \dots + (2n - 1)^2 = \frac{n(2n + 1)(2n - 1)}{3}$$
7

OR
2. a) In a survey of 60 people, it was found that; 25 read Newsweek Magazine, 26 read Fortune, 26 read Time, 9 read both Newsweek and Fortune, 11 read both Newsweek and Time, 8 read both Time and Fortune, 3 read all three Magazines.
 - i) Find the number of people who read atleast one of these magazines.
 - ii) Find the number of people who read exactly one magazine. **7**
- b) State and prove the multiplication principle of counting with an example. **6**
- c) How many different seven-person committees can be formed, each containing 3 women from an available set of 20 women and 4 men from an available set of 30 men ? **7**
3. a) Let $X = \{1, 2, 3, \dots, 7\}$ and $R = \{(x, y) / (x - y) \text{ is divisible by } 3\}$. Prove that R is an equivalence relation and draw the relation graph. **6**
 - b) Let set $A = \{1, 2, 3, 4, 5\}$ and R be relation on A.
 $VERT = [6, 2, 8, 7, 10]$
 $TAIL = [2, 2, 2, 2, 1, 1, 4, 3, 4, 5]$
 $HEAD = [6, 2, 8, 7, 10]$
 $NEXT = [3, 1, 4, 0, 0, 5, 9, 0, 0, 0]$
 - i) Compute digraph of R and M_R .
 - ii) Compute M_R^2 and draw its digraph. **8**

P.T.O.



c) Let $A = \{1, 2, 3, 4, 5, 6\}$ and

$$P_1 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 1 & 2 & 6 & 5 \end{pmatrix}$$

$$P_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 1 & 5 & 4 & 6 \end{pmatrix}$$

Compute :

i) P_1^{-1}, P_2^{-1}

ii) $(P_2 \circ P_1) \circ P_2$.

6

OR

4. a) Explain an equivalence relation with all the properties with an example.

6

b) Let $A = \{1, 2, 3, 4\}$, for the relation R , whose matrix is given, find the matrix of the transitive closure by using Warshall's algorithm.

$$M_R = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

8

c) If $f : A \rightarrow B$ is an everywhere defined function and $|A| = |B| = \text{Finite numbers}$. Then, prove that ' f ' is one-to-one iff ' f ' is onto.

6

5. a) Backtrack to find an explicit formula for the sequence defined by the recurrence relation ; $b_n = 2b_{n-1} + 1$ with the initial-condition $b_1 = 7$.

7

b) Solve the recurrence relation ;

$$d_n = 2d_{n-1} - d_{n-2} \text{ with the initial condition } d_1 = 1.5 \text{ and } d_2 = 3.$$

7

c) Discuss the applications of recurrence relations.

6

OR

6. a) Define recurrence relations with its types.

6

b) Solve $a_n = a_{n-1} + n$ with $a_0 = 4$ using recurrence relation.

7

c) Solve : $a_n = a_{n-1} + n^2$ where $a_0 = 7$ by substitution method.

7



7. a) If 'G' is a connected graph and every vertex has even degree, then there is an Euler circuit in G.

6

b) Write the in-order, pre-order of the tree, whose digraph is shown in Fig. 7 (b).

7

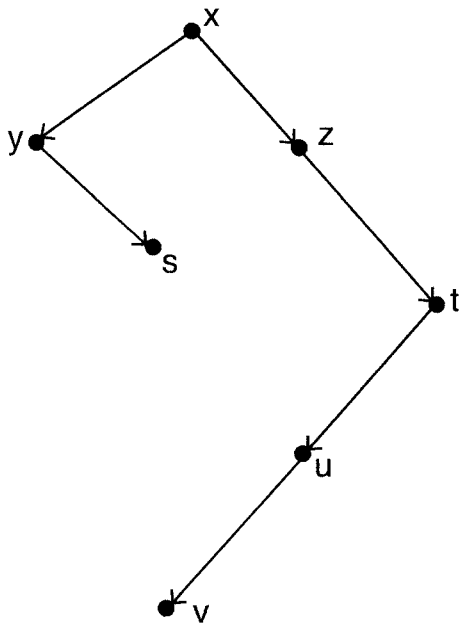


Fig. 7 (b)

c) Define a spanning tree. Find a minimal spanning tree of the weighted graph shown in Fig. 7 (c).

7

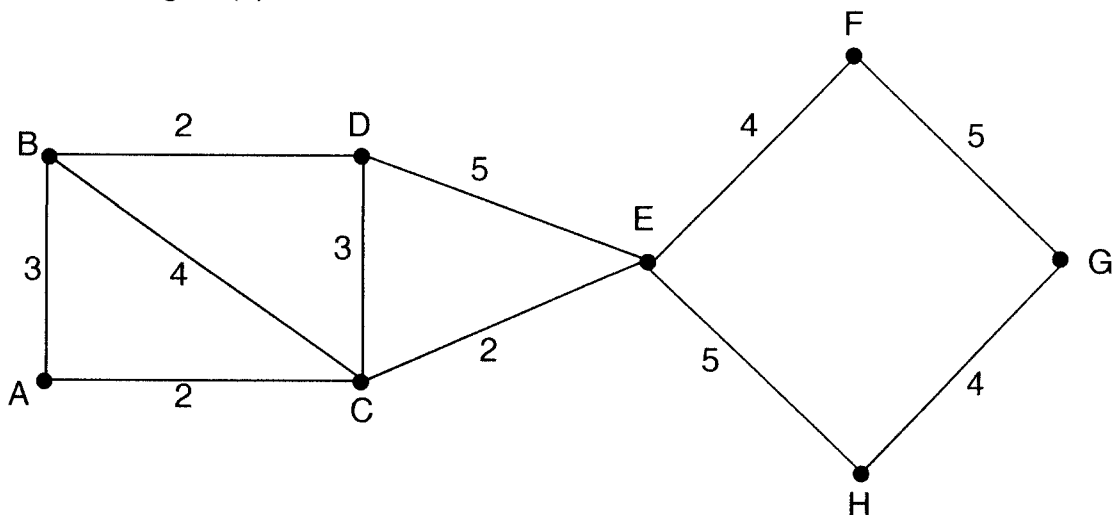


Fig. 7 (c)

OR



8. a) Explain Fleury's Algo. for the given graph and Fig. 8 (a) is it possible to construct an Euler circuit using Fleury's algorithm. If 'YES' and construct the Euler circuit, If 'NO' give explanation.

7

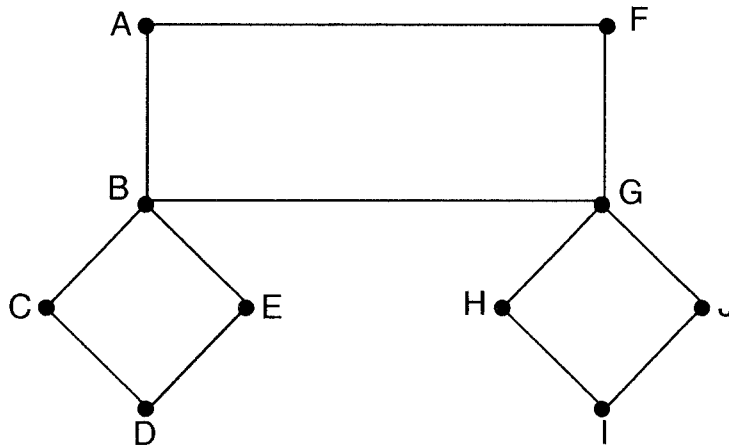


Fig. 8 (a)

- b) Define Hamiltonian path and circuit with an example. 6
- c) Evaluate the following expression and construct the tree of the following algebraic expression, where $x = 5$. 7

$$((2 * x) + (3 - (4 * x))) + (x - (3 * 11)).$$

9. a) Determine the Hasse diagram of the relation on $A = \{1, 2, 3, 4, 5\}$ whose matrix M_R is : 7

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- b) Show that the lattice Z^+ under the usual partial order $\leq$ is distributive. 6
- c) Explain Abelian group with an example. 7

OR

10. a) Define GLB, LUB, Lattice, Distributive and Complemented lattice with an example each. 7
- b) Let $(S, *)$ and $(T, *')$ be semigroups. Show that $S \times T$ and $T \times S$ are isomorphic semigroups. 6
- c) What are idempotent ? Give example. 7



EN – 022

**V Semester B.Tech. (CSE & AIML) Degree Examination, July/August 2024
(NEP Scheme)**

21CCPC 502 : COMPUTER NETWORKS

Time : 3 Hours

Max. Marks : 100

Instruction : Answer questions Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.

1. a) What is a network ? What are the benefits of the networks ? Explain the different categories of network. 10
- b) Explain the OSI reference model with a neat diagram. 10

OR

2. a) What are some common modulation techniques used in the physical layer ? Explain in detail. 10
- b) Compare and contrast between simplex, half-duplex and full-duplex communication. 10
3. a) Define framing. Explain in brief about different framing methods. 10
- b) Given a data stream 1011101 and a divisor 1011, perform CRC (Cyclic Redundancy Check) to detect errors in transmission. 10

OR

4. a) What is Ethernet and how does it operate at the Data Link Layer ? 10
- b) Explain the difference between stop-and-wait and sliding window flow control mechanisms used in the Data Link Layer. 10
5. a) Compare the following : 10
 - i) Hub and Switch
 - ii) Router and Bridge
 - iii) Router and Switch
 - iv) Hub and Bridge.
- b) Given a network with subnet mask 255.255.255.0, calculate the number of host addresses available in the subnet. 10

OR

P.T.O.



6. a) Write short note on :
a) IPV4 addressing
b) IPV6 addressing
c) Routing table. **10**
- b) Given the CIDR (Classless Inter-Domain Routing) notation 192.168.10.0/24, determine the network address, subnet mask and range of host addresses available in this subnet. **10**
7. a) Describe the process of connection establishment and handshaking in TCP and compare with UDP. **10**
- b) Given a TCP segment with sequence number 1000 and acknowledgement number 1500, explain how these numbers are used to ensure ordered and reliable delivery of data. **10**
- OR
8. a) Explain the process of congestion control mechanism in the context of the transport layer. **10**
- b) Draw the header format of UDP and discuss its role in the internet. **10**
9. a) Write short note on :
i) DNS
ii) HTTP
iii) Telnet
iv) FTP
v) SNMP. **10**
- b) How does peer-to-peer file sharing operate at the application layer ? **10**
- OR
10. a) Discuss the primary functions of the application layer in the OSI model. **10**
- b) What is DHCP and how does it function in the application layer ? **10**
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EN – 023

**V Semester B.Tech. (CSE and AIML) Degree Examination, July/August 2024
(NEP Scheme)**

21CCPC503 : DATABASE MANAGEMENT SYSTEMS

Time : 3 Hours

Max. Marks : 100

Instruction : Answer **any five** questions choosing at least **one** from **each** Unit.

1. a) Define DBMS. List and briefly explain the characteristics of Database Approach. 6
- b) With a neat diagram, explain the Three Schema Architecture. 7
- c) Draw an ER diagram for company database by considering at least 4 entities. 7

OR

2. a) What are the different types of database end users ? Discuss the main activities of each. 8
- b) What is an attribute ? Explain the different types attributes used in DBMS with example. 7
- c) Discuss the classification of database languages. 5
3. a) What is a relationship type ? Explain the differences among a relationship instance, a relationship type, and a relationship set. 6
- b) List the operations of relational algebra and the purpose of each with example. 8
- c) Consider the following COMPANY database 6
 EMP (Name, SSN, Salary, SuperSSN, Gender, Dno)
 DEPT(DNum, Dname, MgrSSN)
 DEPT_LOC(Dnum, Dlocation)
 DEPENDENT(ESSN, Dep_name, Sex)
 WORKS_ON(ESSN, Pno, Hours)
 PROJECT(Pname, Pnumber, Plocation, Dnum)

Write the relational algebra queries for the following :

- i) Retrieve the name, address, salary of employees who work for the Research department and whose salary greater than 50000.
- ii) Find the names of employees who work on all projects controlled by department number 4.
- iii) Retrieve the names of employees who have no dependents.

OR

P.T.O.



4. a) What are the Commands used in Data Definition Language in SQL ? Explain each with their syntax and examples. **6**
- b) What are the different types of constraints are available in SQL ? Explain with examples. **6**
- c) Specify the following queries on the COMPANY database schema as shown below and also show the result of each query : **8**
- EMP(Name, SSN, Salary, SuperSSN, Gender, Dno)
 DEPT(DNum, Dname, MgrSSN)
 DEPT_LOC(Dnum, Dlocation)
 DEPENDENT(ESSN, Dep_name, Sex)
 WORKS_ON(ESSN, Pno, Hours)
 PROJECT(Pname, Pnumber, Plocation, Dnum)
- i) Retrieve the names of all employees in department 5 who work more than 10 hours per week on the ProductX project.
- ii) Retrieve the names of all employees who work on every project.
- iii) Retrieve the average salary of all female employees.
- iv) For each project, list the project name and the total hours per week (by all employees) spent on that project.
5. a) Define Normalization. Explain about 1NF, 2NF and 3NF with relevant examples. **10**
- b) Define Functional Dependencies and Discuss about different functional dependencies. **5**
- c) Discuss the generation of spurious tuples. **5**
- OR
6. a) Explain insertion, deletion and updating anomalies with example. **10**
- b) Discuss the join dependencies with an example and explain 5NF. **10**
7. a) What is PL/SQL ? Explain the features of PL/SQL and the basic structure of PL/SQL block with example. **7**
- b) Explain the broad classification of PL/SQL datatypes in detail. **7**
- c) Explain Inheritance in PL/SQL with example. **6**
- OR
8. a) How do we use INSERT, DELETE, UPDATE and SELECT commands used in PL/SQL ? Give example for each command. **8**
- b) Explain the various conditional and looping statements used in PL/SQL with examples. **6**
- c) Write a program in PL/SQL to show the uses of SQL%FOUND to determine if a DELETE statement affected any rows. **6**



9. a) List and discuss the ACID properties of a database transaction. **10**
- b) What is concurrency control and why is it needed explain in detail ? Mention the types of problems faced in concurrency control. **10**
- OR
10. a) How to deal with deadlock detection and starvation ? **7**
- b) Discuss the problems of deadlock and starvation, and the different approaches to dealing with these problems. **7**
- c) Write note on B-trees data structure in file organization. **6**
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EN – 024

**V Semester B.Tech. (CSE & AIML) Degree Examination, July/August 2024
(NEP Scheme)**

21CIPC504 : AUTOMATA THEORY AND COMPILER DESIGN

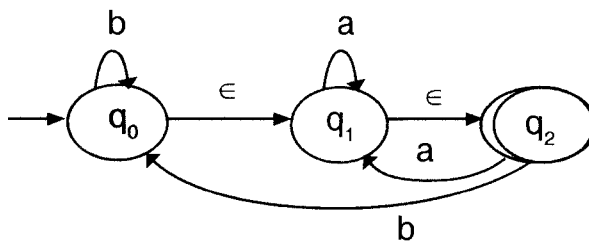
Time : 3 Hours

Max. Marks : 100

Instruction : Answer **any five full** questions, choosing atleast **one full** question from **each** Unit.

UNIT – I

1. a) Define FA. Obtain a DFA to accept strings of a's and b's starting with the string $ab(a+b)^*$ with transition table. 6
- b) Define grammar and explain different types of grammar. 7
- c) Convert the following NFA with ϵ moves to DFA. 7



OR

2. a) Define CFG. Construct the DFA which accepts only those strings representing zero modulo five. 8
- b) Convert the following NFA to DFA. 7

δ_D	0	1
$\rightarrow P$	{P, Q}	{P}
Q	{R}	{R}
R	{S}	$\emptyset$
*S	{S}	{S}

- c) Consider the grammar $S \rightarrow aS | aSbS | E$ 5

Show that deviation for the string aab is ambiguous

P.T.O.



UNIT – II

3. a) Explain LEX Tool with the structure of Lex Program. 7
b) Perform left factor for the grammar 7
 $A \rightarrow abB/aB/cdg/cdeB/cdfB$
c) Describe the role of Compiler. 6
- OR
4. a) What is a Parser ? Explain with a neat diagram different phases of 7
compilers.
b) Explain left recursion and left factoring. 6
c) Explain the concept of input buffering in compiler design. 7

UNIT – III

5. a) Consider the grammar $E \rightarrow E+T/T$, $T \rightarrow T^*F/F$, $F \rightarrow (E)|id$. Design predictive 10
parsing table and check given grammar is LL(1) Grammar or not ?
b) Write a regular expression for relation operators and design a transition 5
diagram for them.
c) Prepare Shift Reduce Parsing for the input string (a,(a, a)) using the grammar 5
 $S \rightarrow (L)|a$ $L \rightarrow L,S|S$.
OR
6. a) Differentiate between shift-reduce and operator precedence parsers. 5
b) Describe the lexical errors and various error recovery strategies with suitable 8
examples.
c) Discuss about error recovery strategies in predictive parsing. 7

UNIT – IV

7. a) Explain in detail about YACC Parser generator tool. 8
b) Define a syntax-directed translation and explain with example. 4
c) Discuss type checking with suitable examples. 8
- OR
8. a) Construct CLR parsing table for the given grammar 10
 $S \rightarrow CC$
 $C \rightarrow aC/d$
b) Define augmented grammar. Explain about handle pruning. 6
c) Differences between SLR, CLR, LALR parsers. 4



UNIT – V

9. a) Define flow graph. Explain structure-preserving transformations for optimization of basic block. 5

b) What is Directed Acyclic Graph (DAG) ? Draw DAG for the following example $t_0 = a + b$; $t_1 = t_0 + c$; $d = t_0 + t_1$. 7

c) i) Translate the expression $A[i] := b$ and $a := B[i]$ into triple representation. 8

ii) Generate the three-address code and compute the basic blocks and draw the flow graph for the following program segment.

```
prod=0;
i=1;
do{
  prod=prod+a[i] * b[i]; i=i+1;
}while(i≤10);
```

OR

10. a) Explain different types of Intermediate Code with an example. 7

b) Construct the DAG for the following basic blocks. 6

- 1) $t1 := 4*i$
 - 2) $t2 := a[t1]$
 - 3) $t3 := 4*i$
 - 4) $t4 := b[t3]$
 - 5) $t5 := t2*t4$
 - 6) $t6 := prod+t5$
 - 7) $prod := t6$
 - 8) $t7 := i+1$
 - 9) $i := t7$
- if $i \leq 20$ goto 1

c) Discuss in detail the role of dead code elimination and strength reduction. 7



EN – 025

**V Semester B.Tech. (CSE and AIML) Degree
Examination, July/August 2024
(NEP Scheme)
21AIPC504 : DIGITAL IMAGE PROCESSING**

Time : 3 Hours

Max. Marks : 100

Instruction : Answer *any five* full questions. **Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.**

1. a) With a neat diagram, explain the fundamental steps in Image Processing. **10**
b) List and explain the applications of Digital Image Processing. **10**

OR

2. a) Discuss the Image sampling and Quantization with an example. **10**
b) List and explain the basic Relationship between pixels. **10**
3. a) Explain the basic Gray Level Transformations used in Image enhancement. **10**
b) Differentiate between Smoothing Spatial Filters and Sharpening Spatial Filters. **10**

OR

4. a) Obtain the Histogram equalization for the following 4 – bit Image Segmentation of 4×4 image. Write the inference on Image Segment before and after Histogram equalization. Also draw the Histogram for Image segment before and after Histogram equalization. **10**

$$\begin{array}{l} \text{I/p} \\ \text{Image} \\ \text{Segment} \end{array} = \begin{bmatrix} 4 & 4 & 3 & 3 \\ 4 & 4 & 3 & 3 \\ 5 & 4 & 4 & 4 \\ 4 & 4 & 4 & 4 \end{bmatrix}$$

- b) Explain how Arithmetic operators are helpful in Image Enhancement. **10**
5. a) Differentiate between Lowpass Filters and Highpass Filters for Image Enhancement in the Frequency Domain. **10**
b) Write a note on Homomorphic Filtering. **10**

OR

P.T.O.



6. a) Discuss the basic steps for Filtering used to Enhance an Image in the Frequency domain. **10**
b) Explain Butterworth and Gaussian Filters with example. **10**
7. a) What is Image Segmentation ? Explain. **10**
b) Explain Three types of Segmentation in Digital Image Processing. **10**

OR

8. a) Explain Prewitt operator, Sobel operator and Laplacian operator with an example. **10**
b) What is Thresholding ? Explain. **10**
9. a) Differentiate between Lossy compression and Lossless compression. **10**
b) Discuss Huffman Coding Technique with suitable example. **10**

OR

10. a) Discuss Image Compression in detail. **10**
b) Explain Arithmetic coding Technique with suitable example. **10**
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EN – 026

V Semester B.Tech. (CSE and AIME) Degree Examination, July/August 2024
(NEP Scheme)
21AIAE 507 : DATA ANALYSIS WITH PYTHON
(Ability Enhancement Course)

Time : 2 Hours

Max. Marks : 100

Instruction : Answer **five full** questions, choosing at least **one** from **each Unit**.

Answer the following. **Each** question carries **20** marks.

1. a) Compare and contrast these four fundamental Python data structures. Discuss when and why a programmer might choose one over the others, including considerations of performance and suitability for different tasks. **10**
- b) Discuss how Python interacts with the operating system to handle file operations. Provide examples of reading from and writing to files, and explain how Python can be used to manipulate file paths, directories, and handle exceptions during these operations. **10**
- OR
2. a) Describe the IPython shell and its advantages over the standard Python shell. What features make IPython particularly useful for data analysis ? **6**
- b) Discuss control flow conditions in Python. **6**
- c) Compare and contrast these four fundamental Python data structures. Discuss when and why a programmer might choose one over the others, including considerations of performance and suitability for different tasks. **8**
3. a) Define universal functions in the context of NumPy. List some unary and binary universal functions with descriptions. **10**
- b) Discuss the various methods available in NumPy for creating ndarrays. Provide examples where you create arrays using at least three different functions, explaining the purpose and output of each. **10**
- OR
4. a) Explain how indexing and slicing work in NumPy arrays. Provide appropriate examples for respective methods. **10**
- b) Illustrate how NumPy performs element-wise operations on arrays. Include examples of operations such as addition, subtraction, multiplication, and division, and explain how broadcasting rules apply. **10**

P.T.O.



5. a) What are the two data structures of Pandas ? Explain with examples. **10**
b) List five indexing options with DataFrames with examples. **10**

OR

6. a) Explain sorting and ranking with respect to Pandas. **10**
b) Explain dropping entries from an axis and re-indexing in both DataFrame and series. **10**
7. a) How to interact with web APIs and databases ? **8**
b) Explain how to read Microsoft Excel files using Python. **7**
c) What are the optional arguments for reading and writing data in text format ? **5**

OR

8. a) List text and binary data loading functions in Pandas with descriptions. **10**
b) How do you read text files in pieces and write data in text format using Python ? **10**
9. a) List five string methods in Pandas with examples. **10**
b) How to handle missing data in preprocessing ? **10**

OR

10. a) How to compute Indicator/Dummy variables in data transforming ? **6**
b) Discuss about extension data types. **4**
c) Explain data transformation in data analysis. **10**
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V Semester B.Tech. (Common to All Branches) Degree
Examination, July/August 2024
(NEP Scheme)

21CVHS508 : ENVIRONMENTAL SCIENCE

Time : 2 Hours

Max. Marks : 100

Instruction : Answer **any five full** questions selecting at least **one full** question from **each** Unit.

UNIT – I

1. a) Explain the scope and importance of environmental studies. 10
b) Which are the main environmental components ? How do they interact with each other ? 10

OR

2. a) Explain the importance of public participation in environmental awareness. 10
b) Discuss the interaction between man and the environment. 10

UNIT – II

3. a) Differentiate between the renewable and non-renewable resources. 10
b) Discuss in detail Chipko Movement. 10

OR

4. a) Explain the impact of mining on forest. 10
b) Define flood. What are the causes of flood ? 10

UNIT – III

5. a) Define ecological pyramids. Explain various types. 10
b) Explain the energy flow in an ecosystem with a flow chart. 10

OR

6. a) Explain producers, consumers and decomposers in a food web and its importance with an example. 10
b) Define ecology. Explain the importance of biotic and abiotic components in an ecosystem. 10

P.T.O.



UNIT – IV

7. a) Explain carbon cycle with a neat sketch. 10
b) Explain ocean ecosystem with different zones. 10

OR

8. a) Explain the structure and function of a forest ecosystem. 10
b) Explain the structure and function of a grassland ecosystem. 10

UNIT – V

9. a) List and explain the causes, effects and control measures of air pollution. 10
b) Explain the noise pollution control measures. 10

OR

10. a) Define solid waste management. How is municipal waste handled in Indian cities ? 10
b) What are the impacts of earthquakes ? How can the effects of earthquakes be mitigates ? 10
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